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9 April 1959
SUPERSEDING
MIL-A-18204
15 March 1955

MILITARY SPECIFICATION

AMMETERS, DIRECT-CURRENT, SPLIT-CORE, PORTABLE

1. SCOPE

1.1 Scope. - This specification covers split-core type ammeters capable of measuring currents in direct-current circuits without circuit interruption.

1.2 Classification. - Split-core ammeters shall be furnished in the following types, as specified (see 6.2):

Type HL - High accuracy, low range.
Type MH - Medium accuracy, high range.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
L-P-349 - Plastic Compounds, Molding, Cellulose Acetate, Butyrate; and Molded or Extruded Parts.
O-L-164 - Leather Dressing, Mildew Preventive.

MILITARY
MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-80 - Plastic Sheet, Acrylic Base, Antielectrostatic, Transparent (for Indicating-Instrument Windows).
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 6625

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3. REQUIREMENTS

3.1 Preproduction sample. - Preproduction samples of direct-current, split-core ammeters, as specified (see 6.2) shall be furnished for preproduction inspection (see 4.2).

3.2 Material. - Material shall be as specified herein. Material not definitely specified shall be of the best quality used for the purpose in commercial practice.

3.2.1 Soldering fluxes. - Noncorrosive soldering fluxes shall be used for all soldering. Excess flux shall be removed.

3.3 Design. - In general, the ammeters shall consist of a hinged split-core and indicating meter movement assembly all mounted in a sturdy, molded plastic case of material conforming to Specification MIL-M-14 or L-P-349, with an integral handle. Ammeter designs such as one consisting of one or more removable molded plastic scale ranges or movement units, and a hinged split-core assembly consisting of partially insulated laminated jaws, a frame that is not insulated, a trigger, hand shield and handle, are allowable. In all designs, the case, handle or any other part that the operator might come in contact with during normal use, shall adequately protect the operator from electrical shock, and shall be free of exposed metal inserts or projections, excluding the split-core, which are not properly insulated. All surfaces of the case and handle shall be smooth and free from flash projections and roughness. The ammeters shall be arranged for current reading by encircling current-carrying conductors of the following sizes:

(a) Type HL. - The ammeter split-core shall be of such size and shape necessary for encircling a minimum of two cables of 1.28 inch diameter.
(b) Type MH. - The ammeter split-core shall be of such size and shape necessary for encircling a minimum of one cable of 2-3/4 inch diameter.

3.3.1 Current ranges. -

3.3.1.1 Type HL. - The ammeter shall have three ranges of 20, 50 and 100 amperes.

3.3.1.2 Type MH. - The ammeter shall have four ranges of 100, 200, 400 and 1000 amperes.

3.3.2 Meter scale. - The meter scales shall be divided by numerical markings in whole number divisions of the full scale rated currents. A numerical division shall represent not less than 2 nor more than 20 percent of the full scale rated currents. The meter shall be protected by means of a heavy cover glass or laminated shatterproof glass or other transparent material free from discoloration scratches, striae, electrostatic effects or other detrimental defects which may interfere in any way with normal reading of the meter in daylight. Plastic windows where used, shall be in accordance with Specification MIL-P-80. The window shall be secured by clamps or other suitable means. Cement alone is not considered sufficient for holding the window in place. When specified (see 6.2), adequate shielding shall be provided to eliminate objectionable effects of stray magnetic fields.

3.3.2.1 Type HL. - The meter scale shall be not less than 1.75 inches in length.

3.3.2.2 Type MH. - The meter scale shall be not less than 2.8 inches in length.

3.3.3 Zero adjuster. - The zero adjuster shall be accessible from the front of the case. The range of adjustment shall be not less than 5 percent of the scale length. The zero adjuster shall be suitably insulated to eliminate all danger to the operator. The zero adjuster assembly shall have sufficient friction to prevent it from shaking loose and changing adjustment when subjected to vibration test specified in 4.7.9.

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3.3.4 Polarity indicator. - A means of determing d.c. polarity shall be furnished in the design of the ammeters. This shall be provided by the addition of an assembly consisting of a switch, battery, coil and associated housing. The coil shall be wound around one section of the split-core and connected through the switch to the battery. The combination shall serve to aid or oppose the magnetic flux introduced into the split-core by the conductor current being measured. Determining cable polarity shall then be possible by noting the direction of meter deflection from the steady state condition. An instruction plate shall be provided for determining polarity.

3.3.5 Accuracy. -

3.3.5.1 Type HL. - The ammeter shall be accurate to within plus or minus 2 percent of full scale deflection when tested in accordance with 4.7.3.

3.3.5.2 Type MH. - The ammeter shall be accurate to within plus or minus 3 percent of full scale deflection on the 100 and 200 ampere ranges, plus or minus 4 percent on the 400 ampere range, and plus or minus 5 percent on the 1000 ampere range when tested in accordance with 4.7.3.

3.3.6 Core. - The core shall be of the laminated type with laminations securely fastened together. The core shall be split type and provided with a pivoted or hinged C- shaped segment, large enough to accommodate a cable or cables having an outside diameter as specified in 3.3. The segment shall be readily operable to both the open and closed positions by means of a light pressure against the conductor or by means of a trigger arranged to open and close the segment.

3.3.7 Range selection. - Range selection shall be accomplished by one of the following:

3.3.7.1 Selector switch. - When a selector switch is used for range selection it shall be clearly marked to permit the user to relate the switch position to the appropriate meter scale. The switch shall be a positive snap action type and located adjacent to the meter handle.

3.3.7.2 Ammeters having interchangeable movement units. - Where interchangeable movement units are used for different ranges, all reference to the ammeter herein shall include the basic assembly plus all movement units.

3.3.7.2.1 Interchangeable movement units. - When interchangeable movement units are used for scale selection they shall be clearly marked with the scale range in accordance with 3.3.2. In addition they shall be clearly marked with the serial number of the meter with which they are to be used.

3.3.8 Carrying case. - Each ammeter shall be provided with a carrying case of either leather, leather substitute, or plywood covered with a tough plastic as specified (see 6.2). The case shall be fitted with carrying handle and lid with a positive fastener, and shall be of adequate size to accommodate the ammeter and interchangeable movement units, if interchangeable movement units are used for different ranges. If of leather, the case shall be treated with a compound conforming to Specification O-L-164 applied in accordance with the recommendations of the manufacturer.

3.3.9 Manual, technical. - Each ammeter shall be provided with an operating technical manual in accordance with type III or IV of Specification MIL-M-15071. Contents shall include equipment description, operation procedure and operating principle

3.4 Physical requirements. -

3.4.1 Weight. - The ammeter, without carrying case and interchangeable movement units (except movement unit installed) shall weigh not more than 5 pounds.

3.4.2 Dimensions. - The overall length of the ammeter without carrying case, shall be not more than 14 inches, overall width not more than 6 inches, and overall height not more than 4 inches.

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3.5 Overload capacity. -

3.5.1 Momentary overload, current circuits. - Each circuit of the ammeter shall be of such design and construction as to withstand, without mechanical or thermal injury, a series of momentary overloads as specified in 4.7.4.1. It shall not be necessary to make any repairs or adjustments, other than resetting the pointer by means of the zero adjuster. The change in accuracy of indication at any point on the scale shall not exceed 2.5 percent of total scale deflection after the ammeter has been conditioned for 1 hour at an ambient temperature of 25° Centigrade (C.) immediately following the application of the overload.

3.5.2 Short-time overload. - The ammeter shall withstand the test specified in 4.7.4.2 without damage to any part or without more than 1.5 percent change in accuracy of indication at any point on the scale. Accuracy shall be determined 20 minutes after the overload test.

3.6 Dielectric strength. -

3.6.1 Circuits, electric and magnetic. - The ammeter shall withstand the test voltage without damage or flashover in the dielectric test specified in 4.7.7.

3.6.2 Zero adjuster. - There shall be no damage to the ammeter or any flashover when the zero adjuster is subjected to a dielectric test voltage of 80 percent of the value specified in 4.7.7.

3.7 Insulation resistance. - The insulation resistance between all the electrical circuits of the ammeter and its case, or any other part which the operator may come in contact with, shall be not less than 20 megohms (see 4.7.8).

3.8 Exposure to extreme temperature. - The ammeter shall operate within the range of plus 65 to 0°C., and shall not be damaged by exposure to temperatures as low as minus 55°C.

3.8.1 High temperature. - The ammeter shall indicate freely at a temperature of 65°C. after having been maintained at this temperature with a relative humidity of not less than 95 percent for 16 hours. The change in accuracy due to operation at this temperature shall not exceed 10 percent when tested as specified in 4.7.5. After the ammeter has been restored to a temperature of 25°C., the permanent error from the beginning of the first cycle to the end of the third cycle, due to heat exposure, shall not exceed 2.5 percent of full scale deflection.

3.8.2 Low temperature. - The ammeter shall indicate freely at a temperature of 0°C., after having first been maintaned at a temperature of minus 55°C. for 2 hours and then maintaned at a temperature 0°C. for 16 hours. The change in accuracy at the latter temperature due to the minus 55°C. exposures shall not exceed 3.0 percent of full scale deflection. After the ammeter has been restored to a temperature of 25°C. the permanent error due to this exposure test shall not exceed 4.0 percent of full-scale deflection. There shall be no increase in the above change in accuracy when the cooling cycle is repeated (see 4 7.6).

3.9 Locking of screw-thread assemblies. - All screw-thread assemblies shall be made vibration-proof. Lock washers or equivalent shall be provided under nuts, where practicable.

3.10 Effect of vibration. - Permanent changes in the accuracy of indications, due to vibration of the character and intensity specified in 4.7.9, shall not exceed 2 percent of full-scale deflection. No screws, bearings, pivots, or other parts shall become loosened or damaged during the test.

3.11 Balance. - When the ammeter is tilted and rotated in any direction, the pointer tip shall remain within 1 degree of the zero point. Moving elements shall show good mechanical balance and minimum effect from positioning in the earth's magnetic field (see 4.7.1).

3.12 Manufacturer's identification. - Each ammeter shall bear the manufacturer's name or trade mark cast integral with, stamped, or otherwise permanently marked upon the item of equipment.

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3.13 Finish. - The ammeter shall be finished in accordance with the best commercial practice for this type of meter.

3.14 Workmanship. - Workmanship shall be in accordance with the best commercial practice for this type of meter. Material shall be free from all defects and imperfections that may affect the serviceability of the finished product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Preproduction inspection. - The preproduction sample shall be submitted to a laboratory designated by the bureau or agency concerned. These samples shall be subjected to the examination and tests specified in 4.6 and 4.7 and any other tests considered necessary to determine conformance with this specification. When a manufacturer can present proof of a previous approval of identical equipment, preproduction inspection may be waived at the discretion of the inspector for the next 2000 ammeters or a period of 5 years whichever occurs first. After this period, the manufacturer will be required to submit a preproduction sample for test.

4.3 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 Sampling. -

4.4.1 Inspection lot. - All ammeters of the same type presented at one time shall be considered a lot for purposes of acceptance inspection.

4.4.2 Sampling for acceptance inspection. - A random sample of ammeters shall be selected in accordance with table I from each inspection lot of material offered for Government inspection in accordance with 4.5. Lot acceptance will be based on sampling inspection requirements in accordance with Standard MIL-STD-105.

Table I - Sampling for visual examination and tests
AQL (approx.) = 1.5 percent defective.

Number of ammeters in inspection lot	Number of ammeters in sample	Acceptance number (defectives)	Rejection number (defectives)
8 and under	2	0	1
9 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 and over	15	0	1

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4.5 Examination and tests. -

4.5.1 Visual and mechanical examination. - Each of the sample ammeters selected in accordance with table I shall be visually and mechanically examined in accordance with 4.6. Any ammeter in the sample containing one or more visual defects shall be rejected, and the lot represented by the sample shall be rejected. Rejected lots may be offered again for inspection provided the contractor has removed all nonconforming ammeters. The inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

4.5.2 Testing for lot acceptance. - Each of the sample ammeters selected in accordance with table I shall be subjected to dielectric strength test and insulation resistance test in accordance with 4.7.7 and 4.7.8. Failure in the dielectric test or insulation resistance test shall be considered failure to meet the requirements of this specification. Each sample should also be tested for accuracy with the ammeter at 25° C. in accordance with 4.7.3 except that no preliminary conditioning is necessary. Each scale shall be checked at 2 points. If the error of any sample ammeter exceeds the specification limits, that ammeter shall be counted as a defective, and shall be rejected, and the lot represented by that sample shall also be rejected. Rejected lots may be offered again for inspeciton provided all ammeters in the lot have been tested by the contractor for the test(s) causing rejection and all noncompliances corrected. Resubmitted ammeters shall be kept separate from new ammeters. Failure of a resubmitted lot to pass any test which caused original rejection will result in final rejection.

4.6 Visual and mechanical examination. - Ammeters shall be examined to verify compliance with 3.2, 3.3 to 3.3.4, 3.4, 3.12 and 3.14.

4.7 Test procedures. -

4.7.1 Balance. - The pointer of the ammeter shall be set to the center of the zero scale division with the ammeter in the horizontal position. The ammeter shall then be tilted approximately 60 degrees in any direction and the position of the pointer noted. The procedure shall be repeated with the ammeter in the vertical position and without resetting the pointer.

4.7.2 Zero adjustment. - The range of the zero adjustment of the ammeter shall be checked for conformance with 3.3.3.

4.7.3 Accuracy. - After preliminary conditioning in a circuit of sufficient capacity for 30 minutes to two-thirds scale deflection the ammeter shall be checked for accuracy at not less than five points (approximately equidistant) distributed over the entrire range of the scale.

4.7.4 Overload capacity. -

4.7.4.1 Momentary overload. - The ammeter shall be subjected to a series of momentary overloads as follows: A load of 3 times the full ampere rating shall be applied to the ammeters on the highest ampere setting nine times for a duration of 1/2 second each, with a 1-minute interval between the successive applications and then followed by one application for a 5-second period (making a total of 10 applications aggregating 9-1/2 seconds during an elapsed period of 9 minutes, 9-1/2 seconds). After 1 hour at an ambient temperature of 25° C., the pointer shall be reset to zero and the calibration checked over the length of the scale.

4.7.4.2 Short-time overload. - A load of 1-1/2 times the full ampere rating shall be applied to the ammeter on the highest ampere setting of the ammeter for a period of 2 minutes (see 3.5.2).

4.7.5 High temperature. - The ammeter shall be conditioned at a temperature of 65° C. with a relative humidity of 95 percent for 16 hours. At the conclusion of this period and while still at this temperature, the meter shall be read at not less than 5 points (approximatly equidstant throughout the calibrated scale). It shall be observed whether or not the meter indicates freely at all points on the scale. The ammeter shall then be allowed to cool to reference temperature for 8 hours and another set of readings taken throughout the calibrated scale. This test shall be made 3 times successively to establish the effect of heat at 65° C. (see 3.6.1).

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4.7.6 Low temperature. - The ammeter shall be conditioned at a temperature of minus 55°C. for 2 hours and then maintained at a temperature of 0°C. for 16 hours. At the conclusion of this period and while still cooled to the latter temperature, the meter shall be read at not less than 5 points, approximately equidistant, throughout the calibrated scale. It shall be observed whether or not the meter indicates freely at all points. The ammeter shall then be restored to the reference temperature for 8 hours and another set of readings taken at the same points. This test shall be made once to establish the effect of exposure at minus 55°C. and 0°C. (see 3.8.2).

4.7.7 Dielectric strength. - The ammeter shall be subjected to a dielectric test voltage of 2200 volts applied between all live parts and the case. The test voltage shall be 60 cycle frequency approximating as closely as possible a true sine wave, and shall be applied continuously for a period of 60 seconds. The measurement of the test voltage shall be made by the voltmeter method, whereby the meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer or through an auxillary ratio transformer. The test shall be made using high-potential test prods having not less than 1/8 inch radius tips. Any damage or flashover shall be noted.

4.7.8 Insulation resistance. - Insulation resistance tests shall be made on the ammeter using a potential of 500 volts direct current; this test shall follow the dielectric strength test as soon as practicable.

4.7.9 Vibration. - The ammeters shall be tested for vibration in accordance with type I of Standard MIL-STD-167. The damage in accuracy due to vibration shall be determined and the ammeter checked for any loosened or damaged parts (see 3.3.3, 3.3.5 and 3.10).

4.7.10 Polarity indication. - The ammeter shall be checked to determine compliance with 3.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Ammeters shall be preserved and packaged by level A or C; packed by level A, B or C and marked as specified (see 6.1) in accordance with Specification MIL-P-17555. Method IIb shall apply for level A preservation and packaging of ammeters.

6. NOTES

6.1 Intended use. - The ammeter is intended for use in measuring current quickly without interrupting the power circuit where extreme accuracy is unnecessary.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2)
(c) Preproduction sample required (see 3.1).
(d) Shielding required (see 3.3.2).
(e) Carrying case material (see 3.3.8).
(f) Selection of level of preservation and packaging and level of packing required (see 5.1).
(g) That: When a manufacturer can present proof of a previous approval of identical ammeters, preproduction inspection may be waived for the next 2,000 units or a period of 5 years, whichever occurs first. After this period the manufacturer is required to submit a preproduction sample for inspection.

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Preparing activity:
Navy - Bureau of Ships
(Project 6625-0052Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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